

Scenario_BiodivChanges

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2025-09-02

Preprocess inputs

Preprocess climate data

The first scenario looks into the impacts of projected climate change on biodiversity. The input data were gathered from CHELSA CMIP6 (SSP585, 2050s) climate inputs (tmax/tmin → tmean) and aggregated from 30° to 1° for MadingleyR compatibility. Data was resampled to a **monthly** temporal resolution, and a spatial resolution of **1 degree**. This resulted in (NUMBER), 12 per year from 2041 and 2070.

```
# Credit: Wenhuan
# ----- A) 1° stack 12 Tmean -----
base_dir <- "/Users/wenhuan/Documents/Climate Data/CHELSA ssp585 2050s"
out_dir <- file.path(base_dir, "madingley_out_ssp585_2050s")
dir.create(out_dir, showWarnings = FALSE, recursive = TRUE)

# 1°
template1deg <- rast(xmin=-180, xmax=180, ymin=-90, ymax=90,
                    resolution=1, crs="EPSG:4326")

# _MM_2041_2070_norm.tif
find_month <- function(mm, var=c("tasmax", "tasmin"),
                      dir=base_dir, period="2041_2070") {
  var <- match.arg(var)
  patt <- sprintf("%s_%s_%s_norm\\.tif$", var, mm, period)
  f <- list.files(dir, pattern=patt, full.names=TRUE, recursive=FALSE)
  if (length(f) != 1)
    stop(sprintf("1 %s_%s %d", var, mm, length(f)))
  f
}

# CHELSA °C
fix_units <- function(r){
  gm <- as.numeric(global(r, "mean", na.rm=TRUE))
  if (!is.na(gm) && gm > 200) r <- r - 273.15 # Kelvin -> °C
  if (!is.na(gm) && gm > 70 && gm < 200) r <- r/10 # °C -> °C
  r
}

# 30° -> 1°
to_1deg <- function(r, target=template1deg){
  fact <- round(1 / res(r)[1]) # ~120 30
  r_agg <- aggregate(r, fact=fact, fun=mean, na.rm=TRUE)
```

```

    resample(r_agg, target, method="bilinear")
}

tmean_1deg_list <- vector("list", 12)

for (m in 1:12) {
  mm <- sprintf("%02d", m)
  fmax <- find_month(mm, "tasmax")
  fmin <- find_month(mm, "tasmin")

  rmax <- fix_units(rast(fmax))
  rmin <- fix_units(rast(fmin))

  # 1°
  rmax_1d <- to_1deg(rmax)
  rmin_1d <- to_1deg(rmin)

  # Tmean
  tmean_1deg_list[[m]] <- (rmax_1d + rmin_1d) / 2
}

tmean_1deg_stack <- rast(tmean_1deg_list)
names(tmean_1deg_stack) <- sprintf("near-surface_temperature_%02d", 1:12)

# Write
for (m in 1:12) {
  writeRaster(tmean_1deg_stack[[m]],
             file.path(out_dir, sprintf("near-surface_temperature_%02d.tif", m)),
             overwrite=TRUE)
}
writeRaster(tmean_1deg_stack,
           file.path(out_dir, "near-surface_temperature_1-12.tif"),
           overwrite=TRUE)

```

Preprocess land cover data

To represent projected human impacts on biodiversity, we used the projected land-use under socio-climatic scenarios from Chen, Li & Liu (<https://www.nature.com/articles/s41597-022-01208-6>). We used data from the SSP5_RCP85 to match the datasets of the Climate change scenarios (see above). The dataset represent land-use in **10 years** intervals, from **2020** to **2100**.

Modifications were made to the dataset to make them MadingleyR-compliant, only the resulting data are provided for size reasons.

- Convert land-use data to HANPP Following Haberl 2007 (<https://www.pnas.org/doi/epdf/10.1073/pnas.0704243104>), we converted the land-use values to HANPP equivalent values. Anthropogenic presence: 0.73 Agricultural land: 0.835 Natural habitat: 1 Everything else: 0 - Resample: The initial data used a 1km resolution, while everything else in MadingleyR used a 1 degree resolution. The dataset was resampled to 1 degree.
- Reprojection: Make sure every raster is in the same projection (4326)

```

library(MadingleyR)
library(terra)
library(tidyverse)

source("R/functions/utils_rasters.R")

input_folder <- "Data/land_cover/land_cover_projection/raw"
output_folder <- "Data/land_cover/land_cover_projection/processed"

# Load MadingleyR default inputs
sptl_inp <- madingley_inputs("spatial inputs")
mask <- sptl_inp[["hanpp"]]

# Read land cover data
r_hanpp_list <- read_rasters(input_folder)

# Process HANPP
for(i in 1:length(r_hanpp_list)){
  r_hanpp <- r_hanpp_list[[i]]

  # Anthropic presence: 0.73 Agricultural land: 0.835
  r_hanpp <- subst(r_hanpp, from=c(5,6), to=c(0.165,0.27))
  r_hanpp[r_hanpp >= 1] <- 1

  # Project to MadingleyR default inputs
  r_hanpp <- project(r_hanpp, mask, method="bilinear")

  r_hanpp_list[[i]] <- r_hanpp

  writeRaster(r_hanpp,
    paste0(output_folder, "/hanpp_",
      names(r_hanpp), '.tif'),
    overwrite=T)
}

```

Run the model

1) Initialize and spin-up

To run the model we loaded the default MadingleyR inputs. We selected the North America region (Canada, United States and Mexico) and crop and masked all the spatial inputs to reduce the size of the files for better handling. We ran the model for an *spin-up* of **200 years** and a *max cohort size* of **100**.

```

library("MadingleyR")
library("terra")
library("sf")
library("rnaturalearth")
library("tidyterra")

set.seed(123)
source("R/functions/utils_rasters.R")

```

```

output_folder <- "~/Desktop/Madingley/output_test"
spatial_window <- c(-170, -50, 15, 83)
countries_list <- c("United States of America",
                    "Canada",
                    "Mexico")
spin_up_years <- 200

if (!dir.exists(output_folder)) {
  dir.create(output_folder, recursive = TRUE)
}

# Load MadingleyR default inputs
sptl_inp <- madingley_inputs("spatial inputs")
chrt_def <- madingley_inputs("cohort definition")
stck_def <- madingley_inputs("stock definition")
mdl_prms <- madingley_inputs("model parameters")

# Get countries shapefile
world <- ne_countries(scale = "medium", returnclass = "sf")
north_america <- vect(world[world$name %in% countries_list, ])

# Crop and mask spatial inputs
sptl_inp <- mask_rasters_with_vector(inputs = sptl_inp,
                                    spatial_vector= north_america)
sptl_inp <- crop_rasters_with_window(inputs = sptl_inp,
                                    spatial_window = spatial_window)
spatial_window <- as.vector(ext(sptl_inp$hanpp)) # Update window

#### Run spin-up ####
# Initialize the model using the pre-loaded inputs
mdata_init <- madingley_init(spatial_window = spatial_window,
                             cohort_def = chrt_def,
                             stock_def = stck_def,
                             spatial_inputs = sptl_inp,
                             max_cohort = 100)
saveRDS(mdata_init, file.path(output_folder, "mdata_init.rds"))

# Run spin-up
dir_path <- paste0(output_folder, "/spin_up")
if (!dir.exists(dir_path)) {
  dir.create(dir_path, recursive = TRUE)
}
mdata_spin_up <- madingley_run(madingley_data = mdata_init,
                              out_dir = dir_path,
                              years = spin_up_years,
                              cohort_def = chrt_def,
                              stock_def = stck_def,
                              spatial_inputs = sptl_inp,
                              model_parameters = mdl_prms,
                              max_cohort = 100)
saveRDS(mdata_spin_up, file.path(output_folder, "mdata_spin_up.rds"))

```

From this point, we use this initial *spin-up* run to implement our different scenarios.

2) Control scenario

To compare the results, we let the model run for the same amount of time the scenarios would run (200 years).

```
control_years <- 200

#### Run control scenario ####
dir_path <- paste0(output_folder, "/control")
if (!dir.exists(dir_path)) {
  dir.create(dir_path, recursive = TRUE)
}
mdata_control <- madingley_run(out_dir = dir_path,
                              years = control_years,
                              madingley_data = mdata_spin_up,
                              spatial_inputs = sptl_inp,
                              silenced = F,
                              max_cohort = 100, # at least 500
                              apply_hanpp = F)
saveRDS(mdata_control, paste0(output_folder, "/mdata_control.rds"))
```

3) Land cover scenario

The HANPP rasters had NULL values in some regions covered by the default spatial data, so we filled the NULL values with 1. Then, we run the model with the HANPP data for 10 years with each raster from 2020 to 2090. With the 2100 raster we run the model for 100 years.

```
hanpp_folder <- "Data/land_cover/land_cover_projection/processed"
steps_years <- 10
final_years <- 100

#### Load spin-up ####
mdata_lc_list <- list()
mdata_lc_list[[1]] <- mdata_spin_up

#### Preprocess HANPP data ####
r_hanpp_list <- read_rasters(hanpp_folder)
# Remove NA's
r_hanpp_list <- lapply(r_hanpp_list, function(r) {
  r[is.na(r[])] <- 1
  r
})
r_hanpp_list <- mask_rasters_with_vector(inputs = r_hanpp_list,
                                       spatial_vector = north_america)
r_hanpp_list <- crop_rasters_with_window(inputs = r_hanpp_list,
                                       spatial_window = spatial_window)

#### Run land cover scenario ####
# Run the Madingley model with HANPP input
for(i in 1:length(r_hanpp_list)){
  r_hanpp <- r_hanpp_list[[i]]
  hanpp_name <- names(r_hanpp)
  print(hanpp_name)
```

```

sptl_inp$hanpp[] <- r_hanpp
plot(sptl_inp$hanpp)

dir_path <- paste0(output_folder, "/", hanpp_name)
if (!dir.exists(dir_path)) {
  dir.create(dir_path, recursive = TRUE)
}

# Run the model for 100 years with the last HANPP raster
# and for 10 years with the rest
if(i == length(r_hanpp_list)){
  n_years <- final_years
} else {
  n_years <- steps_years
}
mdata_lc_list[[i+1]] <- madingley_run(years = n_years,
                                     out_dir = dir_path,
                                     madingley_data = mdata_lc_list[[i]],
                                     spatial_inputs = sptl_inp,
                                     silenced = F,
                                     max_cohort = 100,
                                     apply_hanpp = 1)
saveRDS(mdata_lc_list[[i+1]], paste0(output_folder, "/", hanpp_name, ".rds"))
}

```

5) Climate change scenario

```

climate_folder <- "Data/madingley_out_ssp585_2050s"
n_years <- 1

#### Load spin-up ####
mdata_climate_list <- list()
mdata_climate_list[[1]] <- mdata_spin_up

# Preprocess temperature data
r_temp <- rast(file.path(climate_folder, "near-surface_temperature_1-12.tif"))
r_temp <- mask_rasters_with_vector(inputs = r_temp,
                                   spatial_vector = north_america)
r_temp <- crop_rasters_with_window(inputs = r_temp,
                                   spatial_window = spatial_window)

#### Run climate change scenario ####
r_temp_control <- sptl_inp$`near-surface_temperature`
sptl_inp$`near-surface_temperature` <- r_temp

dir_path <- paste0(output_folder, "/climate_scenario")
if (!dir.exists(dir_path)) {
  dir.create(dir_path, recursive = TRUE)
}
mdata_climate_list[[2]] <- madingley_run(years = n_years,
                                         out_dir = dir_path,

```

```

        madingley_data = mdata_climate_list[[1]],
        spatial_inputs = spt1_inp,
        silenced = F,
        max_cohort = 100,
        apply_hanpp = F)

saveRDS(mdata_climate_list[[2]], paste0(output_folder, "/climate_scenario.rds"))

```

4) Remove large herbivores

To represent extinction events, we removed certain cohorts. In this case, all the large herbivores were removed. We reproduced the tests made by (<https://madingleyr.github.io/MadingleyR/CaseStudies/CASESTUDY1.html>).

Get Simran's code here

Scenario: Remove large carnivores

To represent extinction events, we removed certain cohorts. In this case, all the large carnivores were removed. This test is inspired by (<https://madingleyr.github.io/MadingleyR/CaseStudies/CASESTUDY1.html>).

```

mdata_carni = mdata2

remove_idx = which(mdata_carni$cohorts$AdultMass > 1e5 & mdata_carni$cohorts$FunctionalGroupIndex ==1)

cohort_defs$PROPERTY_Maximum.mass[2] = 100000

cohort_defs=madingley_inputs('cohort definition')

```